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Principios de Diseño Orientado a Objetos para Ingenieros Senior & Staff

1. Resumen Técnico de Principios SOLID

- **S - Single Responsibility Principle (SRP):** Una clase debe tener una y solo una razón para cambiar.
- **O - Open/Closed Principle (OCP):** Abierto para extensión, cerrado para modificación.
- **L - Liskov Substitution Principle (LSP):** Subtipos deben ser sustituibles por sus tipos base sin alterar el comportamiento.
- **I - Interface Segregation Principle (ISP):** Es mejor tener interfaces específicas que una interfaz de propósito general gigante.
- **D - Dependency Inversion Principle (DIP):** Depende de abstracciones, no de concreciones.

2. Demostración en Código C# (.NET 8)

```
// Bad: Rompe SRP e ISP
public interface IOrderProcessor {
    void ProcessOrder(Order order);
    void SaveToDatabase(Order order);
    void SendEmailNotification(Order order);
}

// Good: Aplicando SRP, ISP y DIP
public interface IOrderProcessor {
    Task ProcessAsync(Order order);
}

public interface IOrderRepository {
    Task SaveAsync(Order order);
}

public interface INotificationService {
    Task SendConfirmationAsync(Order order);
}
```

Simulated Interview Response (English - Senior / Staff Level)

Interviewer: "How do you apply SOLID principles when designing enterprise applications in .NET?"

Your Answer:

"In my everyday work as a Senior Engineer, SOLID principles aren't just theoretical guidelines—they dictate how I structure code for maintainability and testability. For example, I heavily enforce the **Single Responsibility Principle** by isolating business logic from infrastructure concerns. I rely on the **Dependency Inversion Principle** by depending on abstractions rather than concrete classes, which seamlessly enables dependency injection and simplifies unit testing with mocks. Additionally, when designing clean interfaces, I keep them lean following the **Interface Segregation Principle** so that clients aren't forced to implement unused methods."